# Yoav Gudes

Semiconductor Engineering Co-op Student
University of Waterloo 2028
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## Summary


- Waterloo Nanotechnology Engineering student focused on semiconductors, with 16 months of co-op experience.
- Hands-on experience with NVIDIA/AMD GPU testing and validation; machine learning, and data pipeline development.
- Strong foundation in semiconductor devices and circuit design, including cleanroom fabrication of advanced capacitors.
- Leadership experience managing teams and student initiatives across campuses.


## Education

**Nanotechnology Engineering:** Nanoelectronics Specialization **|** **University of Waterloo** **Sept 2023 – Apr 2028**

## Work Experience

**Process Engineering Intern | WOLF Advanced Technology** **Sept 2025 – Apr 2026**

- Conducted GPU stack-up failure analysis across 10 units, identified root causes and supported design improvements.
- Executed hands-on assembly and diagnostics of 50+ NVIDIA and AMD GPU systems via electrical and environmental testing.
- Designed and validated thermal solutions in SolidWorks, including heat pipe and vapor chamber assemblies.
- Authored engineering reports evaluating MIL-STD compliance and translating requirements into design criteria.

**AI/ML Engineering & Data Science Intern | TD Bank** **Jan 2025 – Apr 2025**

- Developed a sentiment analysis model using XGBoost, achieving 90% accuracy on client call classification.
- Built a supervised learning dataset of 10,000+ labeled utterances to train and optimize model performance.

**Data Engineering Intern | MedReddie** **May 2024 – Aug 2024**

- Developed an AI-driven pipeline for automated collection and structuring of data for 8,000+ suppliers and $50M+ products.
- Automated 1,000+ DDQs using LLM APIs, improving AI-driven product recommendation accuracy.

## Technical Skills

**Semiconductor Devices & Circuit Design**

- *Device Analysis:* Diodes, MIS Capacitors & MOSFETs
- *Analog Circuits:* Biasing, Small-Signal Analysis & Amplifiers
- *Digital Circuits:* Transistor Switching & CMOS Logic
- *Optoelectronics:* LEDs, Lasers & Photodetectors

**Semiconductor Fabrication & Processing**

- *Thin-Film Deposition:* PECVD & DC Sputtering
- *Photolithography:* Alignment, UV Exposure & Development
- *Etching:* Reactive Ion Etching & Wet Chemical Etching

**Materials & Electrical Characterization**

- *Spectroscopy:* FTIR, Raman & UV–Vis
- *Electron Microscopy:* SEM & TEM
- *Structural Analysis:* Ellipsometry & X-ray Diffraction
- *Electrical Testing:* C–V/I–V Analysis & 4-Point Probe

**Software, Data & Machine Learning**

- Python, MATLAB & SolidWorks
- Model Training & Optimization
- Data Processing & Pipeline Development

## Leadership & Achievements

**Student Body President | Hillel Waterloo & Laurier** **May 2025 – Aug 2026**

- Led student programming and oversaw executive board operations across a multi-campus organization.
- Addressed student concerns through advocacy, campus initiatives, and direct engagement.

**Awards**

- **University of Waterloo:** Term Distinction (Dean's List), Excellent Academic Standing & Co-op Review, Jordan Banks Award.
- **TD Bank:** Achieve the Dream Canada Scholarship, Co-op Challenge Finalist
- **Hillel Ontario:** Lewis and Olga Boros Scholarship

**Languages**

- English & Hebrew (Fluent)**;** Portuguese & Spanish (Conversational)

**References Available Upon Request**